

EXP NO 3	SETTING UP HYBRID CLOUD CONNECTIVITY USING CISCO PACKET TRACER
Date: 22.07.2026	

Aim

To design and simulate a hybrid cloud network topology connecting a private on-premises network with a public cloud environment using Cisco Packet Tracer, and to verify connectivity between the private network and the simulated cloud.

Introduction to Hybrid Cloud

Hybrid Cloud is a cloud computing model that combines a private, on-premises infrastructure with a public cloud, allowing data and applications to be shared between them. It gives organizations the flexibility to keep sensitive workloads on the private network while using the public cloud for scalability, backup, or additional compute resources.

Cisco Packet Tracer is a network simulation tool that allows students and administrators to design, configure, and test network topologies, including routers, switches, end devices, and a built-in Cloud component that can be used to represent connectivity to an external / public cloud network.

A hybrid cloud connectivity setup typically involves:

- A private network segment (LAN) with end devices, a switch, and a router
- A router configured with Network Address Translation (NAT) or a routed WAN link
- A Cloud component representing the public cloud network
- Routing / NAT configuration to allow communication between the two networks

Software / Environment Used

- Simulation Tool: Cisco Packet Tracer
- Devices: Router, Switch, PCs, Cloud component
- Protocols: IP Addressing, Static Routing, NAT

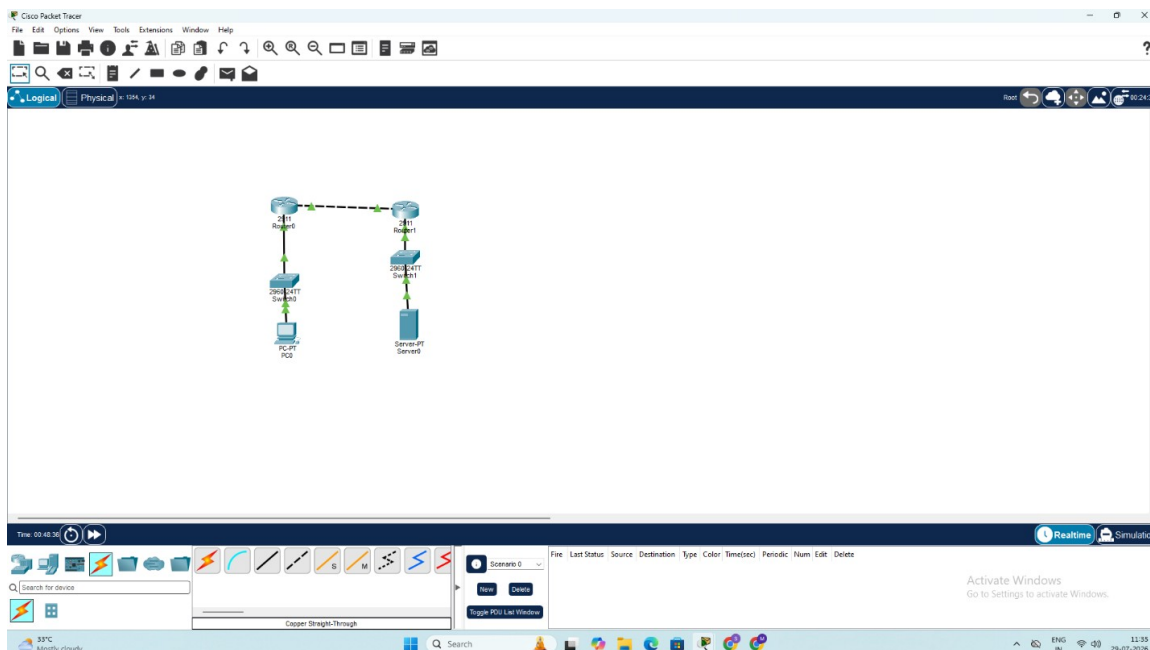
Procedure

Step 1: Launch Cisco Packet Tracer and Create a New Project

Cisco Packet Tracer was launched and a new blank workspace was created for designing the hybrid cloud topology.

Step 2: Build the Private (On-Premises) Network

A router and a switch were placed on the workspace, and two or more PCs were connected to the switch to represent the private LAN. All devices were connected using the appropriate copper straight-through and crossover cables.



Step 3: Add the Cloud Component

A Cloud component was added from the WAN Emulation category to represent the public cloud network, and it was connected to the WAN-facing interface of the router using a serial or Ethernet link, as supported by Packet Tracer.

Step 4: Configure IP Addressing on the Private Network

Static IP addresses were assigned to the PCs and to the LAN interface of the router. The default gateway on each PC was set to the router's LAN interface address.

PC IP: 192.168.1.10 /24 Gateway: 192.168.1.1

Router LAN Interface: 192.168.1.1 /24

Step 5: Configure the WAN Link to the Cloud

The router's WAN-facing interface connected to the Cloud component was configured with a public-style IP address, and the interface was enabled using the no shutdown command.

```
Router(config)# interface <wan-interface>
```

```
Router(config-if)# ip address <public-ip> <subnet-mask>
Router(config-if)# no shutdown
```

Step 6: Configure Routing / NAT

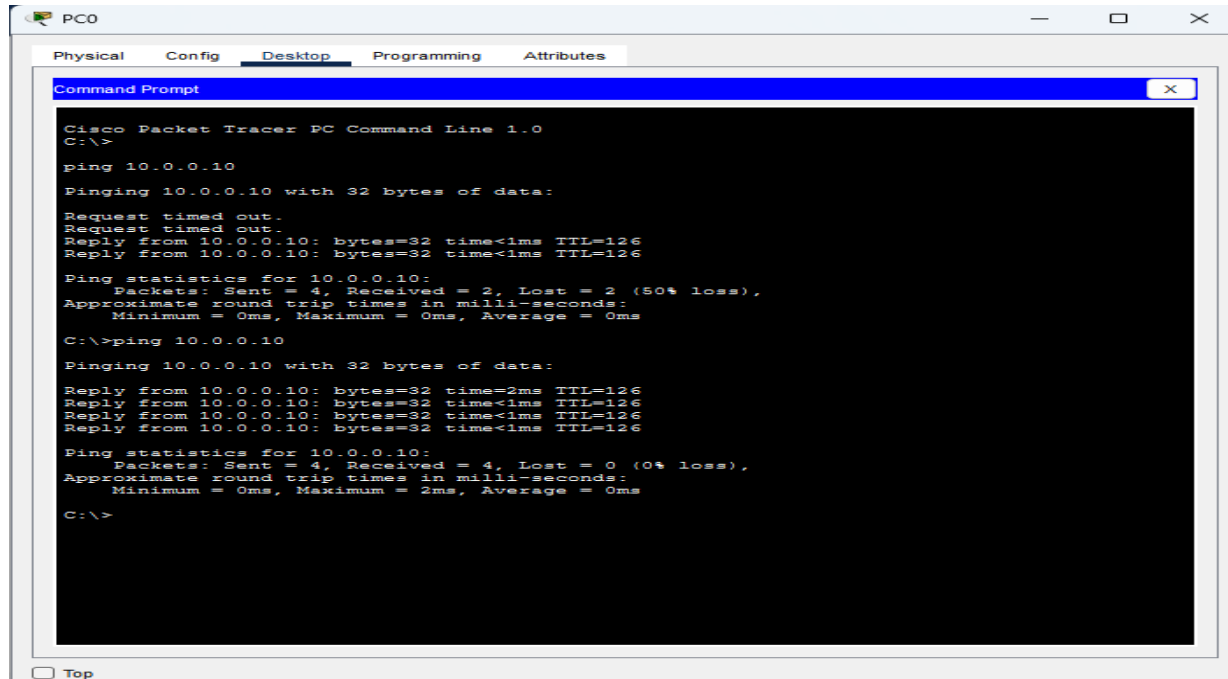
A static default route (or NAT overload configuration) was added on the router so that traffic from the private LAN could reach the cloud-side network through the WAN interface.

```
Router(config)# ip route 0.0.0.0 0.0.0.0 <wan-interface>
```

Step 7: Verify Hybrid Cloud Connectivity

Connectivity between a private-network PC and the cloud-side endpoint was verified using the ping command from the PC command prompt. A successful reply confirmed that the private on-premises network was able to communicate with the simulated public cloud.

```
C:\> ping <cloud-endpoint-ip>
```



Conclusion

A hybrid cloud network topology connecting a private on-premises network with a simulated public cloud was successfully designed and configured using Cisco Packet Tracer. IP addressing, routing, and connectivity between the private network and the cloud component were verified, thereby meeting the objectives of the experiment.